

Hess's law

Learning outcomes

By the end of this section you should be able to apply Hess's law to calculate enthalpy changes in multistep reactions.

Imagine you are going for a hike to the top of a mountain and have a choice of paths to take you there. You could go straight up the mountain or take a side route that brings you to a lake or maybe a viewpoint. Whichever path you choose, you end up in the same place (**Figure 1**).



Figure 1. Many paths up a mountain all leading to the same place.

Credit: TravelCouples, Getty Images

Like hiking to the top of a mountain, the net enthalpy of a chemical reaction is independent of the path taken, therefore, enthalpy is a state function. A state function is defined as a property whose value does not depend on the path taken to reach that value. Other examples of state functions that you will have come across in the DP chemistry course are temperature, pressure and volume.

Nature of Science

Aspect: Experiments

Scientists design and perform experiments to obtain data, which can be used to test hypotheses.

Hess's law calculates the enthalpy change for a reaction indirectly through other reactions, rather than directly using experimental evidence for the reaction itself. When might this be a benefit?

Hess's law, published by Germain Henri Hess in 1840, states that the total enthalpy change in a chemical reaction is independent of the route by which the chemical reaction takes place, as long as the initial and final conditions are the same. This means that we can calculate the enthalpy change of a chemical reaction by considering different possible steps to reach the same products, much like the different pathways that reach the top of the mountain. Hess's law is an application of the law of conservation of energy since both state that energy is conserved.

Concept

Hess's law states that the total enthalpy change in a chemical reaction is independent of the route by which the chemical reaction takes place, as long as the initial and final conditions are the same.

Energy cycles

Energy cycles provide us with a useful tool when applying Hess's law in determining the enthalpy of a reaction. **Figure 2** shows two separate pathways from reactants to products. Both achieve the same product so both pathways could be used to determine the overall enthalpy change for the reaction.

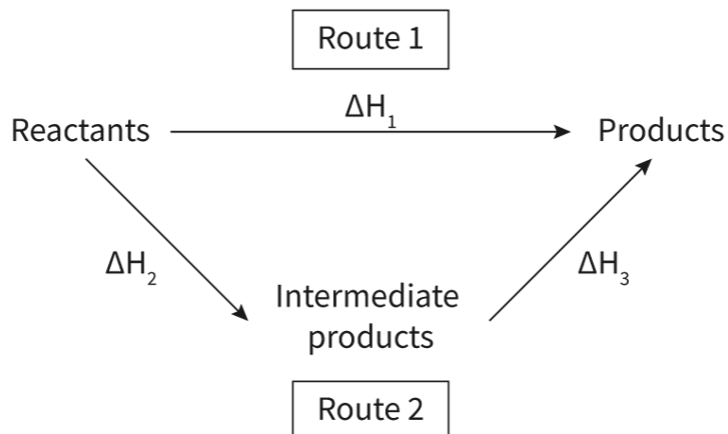


Figure 2. Energy cycle showing two separate pathways to reach the products.

🕒 More information for figure 2

The overall enthalpy of the reaction in **Figure 2** could be determined in two ways:

1. Enthalpy of the direct route (**Route 1**) = ΔH_1
2. Enthalpy of the indirect route (**Route 2**) = $\Delta H_2 + \Delta H_3$

From this we can conclude that $\Delta H_1 = \Delta H_2 + \Delta H_3$

Importantly, you can solve for any of the enthalpies in the equation by rearranging the terms:

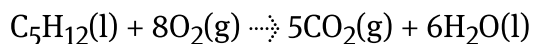
$$\Delta H_2 = \Delta H_1 - \Delta H_3 \quad \text{and} \quad \Delta H_3 = -\Delta H_2 + \Delta H_1$$

Note that if you go against an arrow in an energy cycle, the value of the enthalpy change is reversed.

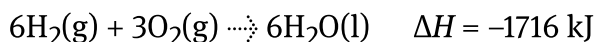
Equation method for Hess's law

Calculating the enthalpy of reaction using Hess's law involves manipulating the different pathway equations and cancelling out those atoms or molecules that occur on both sides of the reaction arrow. This method can be thought of as proof that energy cycles work.

Consider the following example for the combustion of pentane:



Using the equations below, together with their known enthalpy changes, the enthalpy change of the above reaction can be calculated.



Step 1: Think of each molecule in the reaction you are solving for as a 'puzzle piece' where you need to move it to the correct spot:

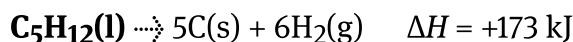
Reaction you are solving for:



'Puzzle pieces':

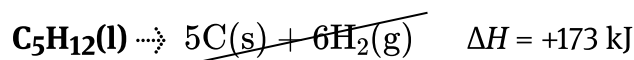


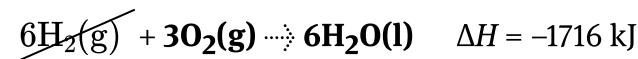
From this, we can see that in our 'puzzle pieces', $\text{C}_5\text{H}_{12}(\text{l})$ is on the wrong side of the reaction arrow, so we reverse the reaction and do the same to the enthalpy change:



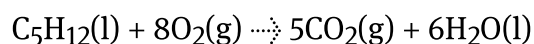
In the second and third equations, the O_2 , CO_2 and H_2O are on the correct side of the arrow and the sum of the coefficients equals the original reaction equation.

Step 2: Cross off all molecules that appear on both sides of the reaction arrow. The $5\text{C}(\text{s})$ and $6\text{H}_2(\text{g})$ on each side of the arrow cancel out.





Step 3: Write the remaining molecules as a chemical equation. If you have set up your equations properly, you will be left with the reaction that you were solving for.



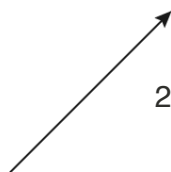
Step 4: Calculate the ΔH for the reaction by adding the enthalpy changes for each individual reaction:

$$\Delta H = +173 \text{ kJ mol}^{-1} + (-1970 \text{ kJ mol}^{-1}) + (-1716 \text{ kJ mol}^{-1}) = -3513 \text{ kJ mol}^{-1}$$

Try the example in **Interactive 1** by dragging and dropping the equations and coefficients so the alternative route is equal to the given reaction.



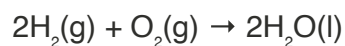
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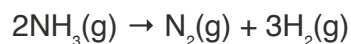
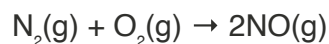
2 ×

×

2



3


☒ Check

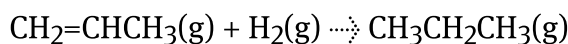
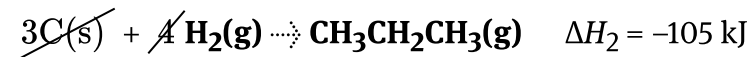
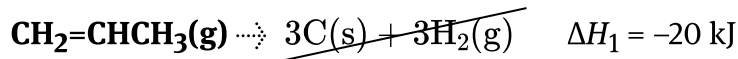
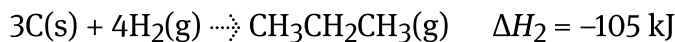
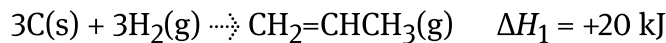
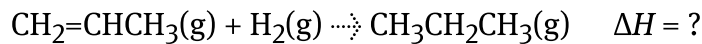
Interactive 1. Hess's law.

Study skills

It is very easy to make errors in Hess's law questions if you are not paying close attention to the states of the chemical species. When you are cancelling out chemical species on either side of the reaction arrow, you can only cancel out those that are in the same state. For example, you can cancel out $\text{H}_2\text{O}(\text{l})$ with $\text{H}_2\text{O}(\text{l})$, but **not** $\text{H}_2\text{O}(\text{g})$ with $\text{H}_2\text{O}(\text{l})$.

Worked example 1

Using Hess's law and the provided equations, calculate the enthalpy of reaction for the hydrogenation of propene using Pd as a catalyst:

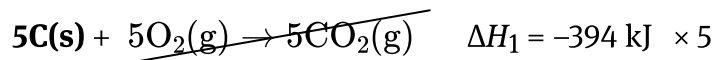
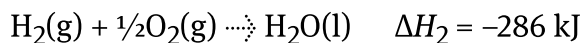
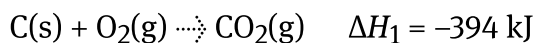
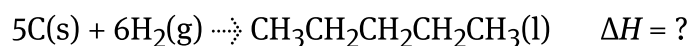


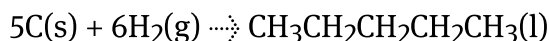
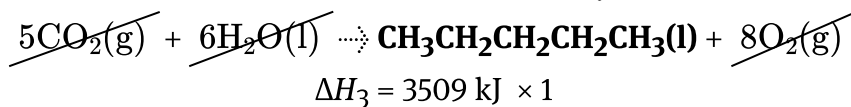
Therefore,

$$\begin{aligned} \Delta H &= (-1)\Delta H_1 + \Delta H_2 \\ &= (-1)(+20) + (-105) \\ &= -125 \text{ kJ mol}^{-1} \end{aligned}$$

Worked example 2

Using the data provided, what is the enthalpy of reaction for the following?

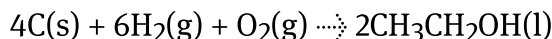
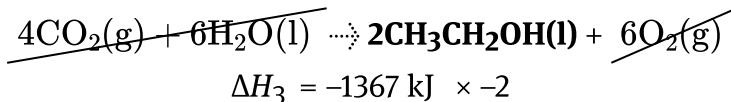
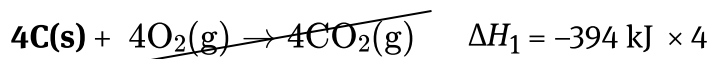
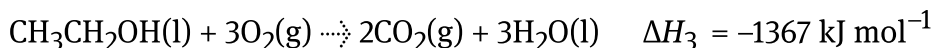
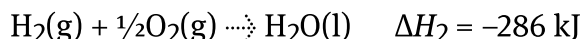
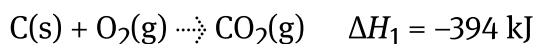
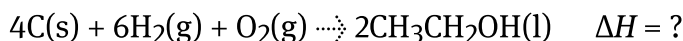




Therefore,

$$\begin{aligned}\Delta H &= 5\Delta H_1 + 6\Delta H_2 - \Delta H_3 \\ &= 5(-394) + 6(-286) + (3509) \\ &= -177 \text{ kJ mol}^{-1}\end{aligned}$$

Worked example 3



Therefore,

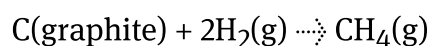
$$\begin{aligned}\Delta H &= 4\Delta H_1 + 6\Delta H_2 + (-2)\Delta H_3 \\ &= (4)(-394) + (6)(-286) - (2)(-1367) \\ &= -558 \text{ kJ mol}^{-1}\end{aligned}$$

Alternate method for Hess's law

Hess's law can also be approached using an energy cycle diagram method. Mathematically, the answer will be the same as the equation method previously presented, but this method provides an illustrative presentation of the equations instead.

Study skills

There is always more than one way to solve a question and your job as a learner is to determine what works best for you. The equation method for answering Hess's law question is a perfect example of this! Keep in mind that you should be able to answer questions using both methods, but when given a choice it is up to you.



Instead of using chemical equations for the separate pathways, a diagram can be used instead. In the diagram shown in **Figure 3**, the equations provided are for the combustions of each chemical species in the equation that you are solving for (the formation of methane).

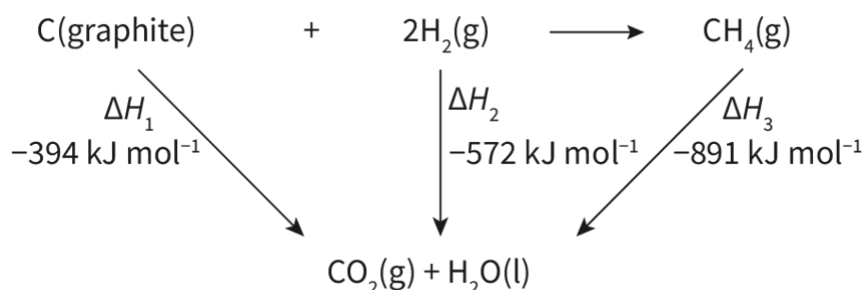


Figure 3: Energy cycle for Hess's law for the formation of methane from its elements.

🕒 More information for figure 3

In this reaction, we can see that two moles of hydrogen gas are required to form one mole of methane. Therefore, we must multiply the enthalpy for hydrogen gas by two:

$$\begin{aligned}\Delta H_2 &= 2 \times -286 \text{ kJ mol}^{-1} \\ &= -572 \text{ kJ mol}^{-1}\end{aligned}$$

To determine the enthalpy of the reaction we will use:

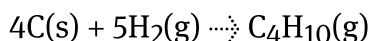
$$\Delta H_r = \Delta H_1 + \Delta H_2 - \Delta H_3$$

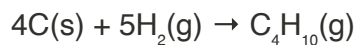
We need to imagine that we are 'flipping the arrow' for ΔH_3 so that it is pointed at the products. For this reason, we need to change the sign to a negative. This was the same strategy that we used for the equation method when we multiplied by a negative number if we need to flip the reaction from reactants to products, to products to reactants instead. If we were to write out the equations, we would see that this allows us to properly cancel out all the chemical species which are not in the overall equation.

Therefore:

$$\begin{aligned}\Delta H_r &= [(-394) + (2 \times -286)] - (-891) \\ &= -75 \text{ kJ mol}^{-1}\end{aligned}$$

Sometimes one of the hardest parts of solving Hess's law questions is setting up the equations. Use **Interactive 2** to arrange the energy cycle to determine the enthalpy of reaction for:





×
-394
KJ mol⁻¹

×
-286
KJ mol⁻¹

×
-2878
KJ mol⁻¹

carbon dioxide + water

ΔH = KJ mol⁻¹

32

1

-1

-32

2

-2

64

3

-3

-64

4

-4

128

5

-5

-128



✓ Check

Interactive 2. Enthalpy of reaction of carbon and hydrogen to give butane.

Creativity, activity, service

Strand: Service

Learning outcome: Demonstrate engagement with issues of global significance

Heat is a form of energy and is defined as the movement of energy from a hotter temperature object to a cooler one. We often feel heat changes in our daily lives but how do changes in heat affect the bigger picture? One such example is global warming and climate change [↗\(https://climate.nasa.gov/global-warming-vs-climate-change/\)](https://climate.nasa.gov/global-warming-vs-climate-change/) which could have devastating effects on the environment.

As an IB student, think about how you can demonstrate engagement with issues of global significance such as these. Remember that small changes made in your local community can lead to big changes on a global scale. 'Think globally, act locally' is a phrase that you should keep in mind when carrying out your CAS experiences.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Thinking skills— Applying key ideas and facts in new contexts
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual/pair activity

How do the states/phases of chemical species impact the enthalpy of reaction and how can we use Hess's law to investigate this?

- Using Hess's law, solve for the enthalpy of reaction for the following:
 - Solid NaOH dissolved in water
 - Aqueous NaOH reacted with aqueous HCl
 - Solid NaOH reacted with aqueous HCl
- Compare and contrast the enthalpies and discuss how they relate to one another.
- Write a conclusion statement that explains the impact of the state of a substance on the overall enthalpy of a reaction with reference to the NaOH/HCl chemical system.